

AutoChecklist

Automated Checklist Refinement for LLM Judges

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THE PROBLEM

Evaluation → **scalable, interpretable, & reliable.**

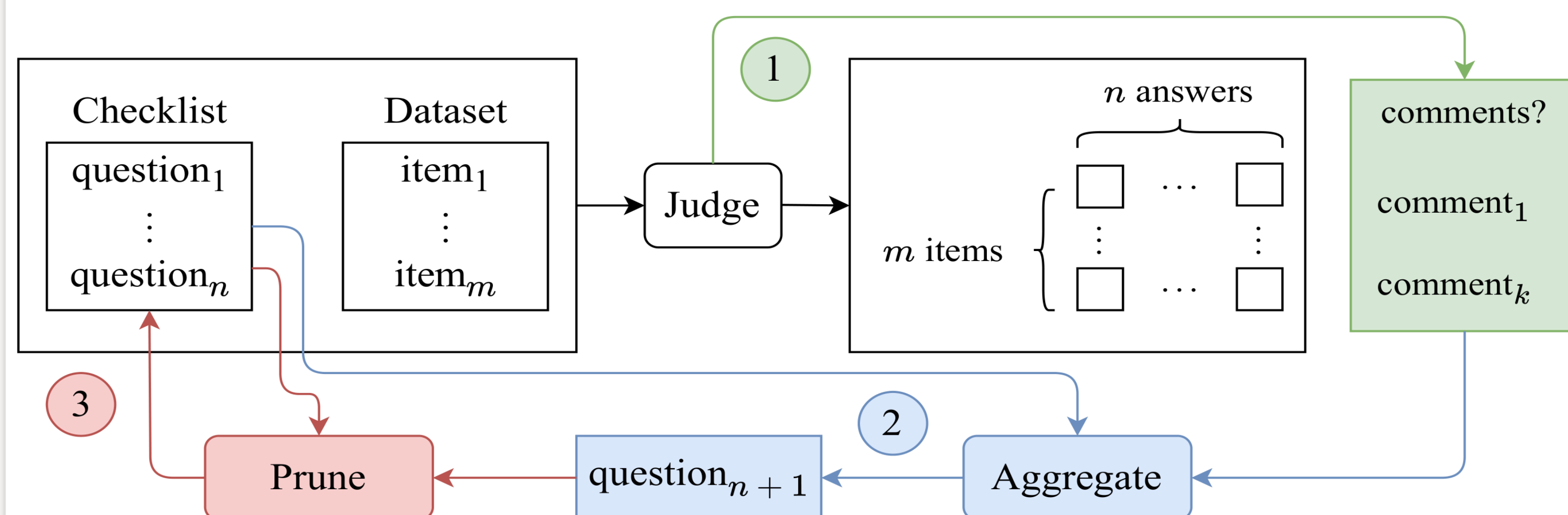
- Metrics like BLEU / ROUGE miss semantic quality.
- Human judgment is costly, slow, inconsistent.
- LLM judges scale but lack reproducibility & transparency.
- **Checklists make judging objective** —
but are hand-built, static, and miss data-specific failures.

THE KEY IDEA

Take free-form comment from human surveys

Give the LLM judge room to flag anything unusual — then turn those comments into new checklist questions, so the checklist evolves with the data it actually sees.

HOW IT WORKS · A SELF-REFINING LOOP



1 Generate: judge answers checklist + free-form comment

2 Aggregate: comments folded into new checklist questions

3 Prune: redundant questions removed: **then loop repeats...**

RQ1 · COMMENTS IMPROVE JUDGMENTS

> 90%

of answers agree with the no-comment run

better

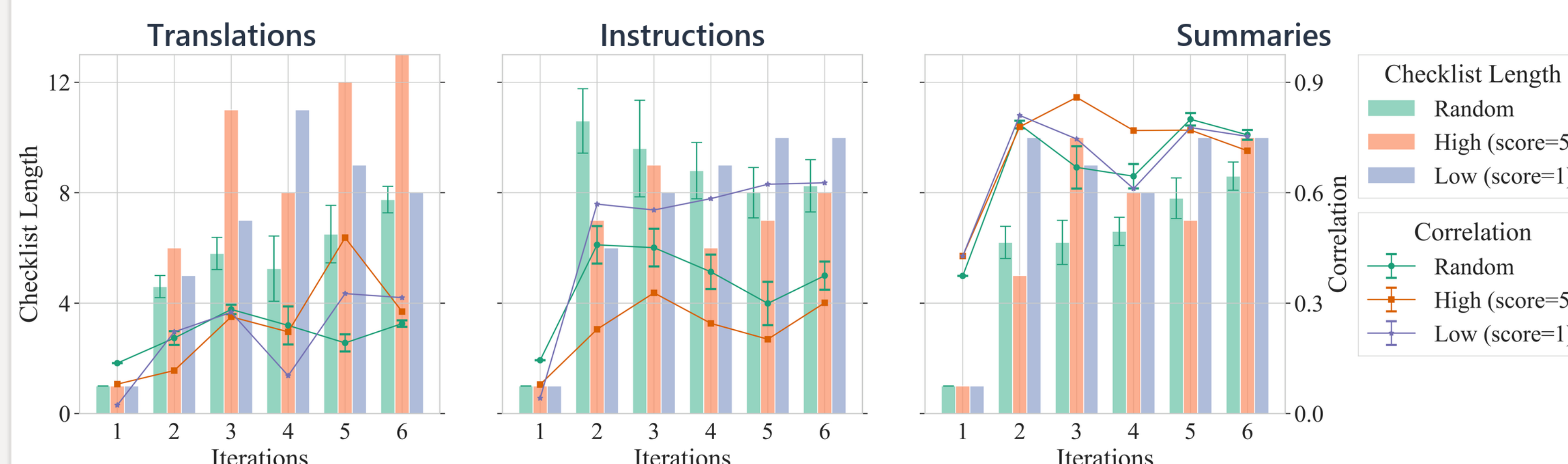
when answers differ, **commented one wins**

No cost

comments never degrade the checklist

Why: for “**Highlight value > 3 on G9:G17**”, the no-comment judge wrongly says “**ambiguous.**”
With a comment, it names the real cause — a **typo**, not ambiguity — so it stops over-clustering unrelated failures.

RQ3 · WHAT YOU REFINE ON



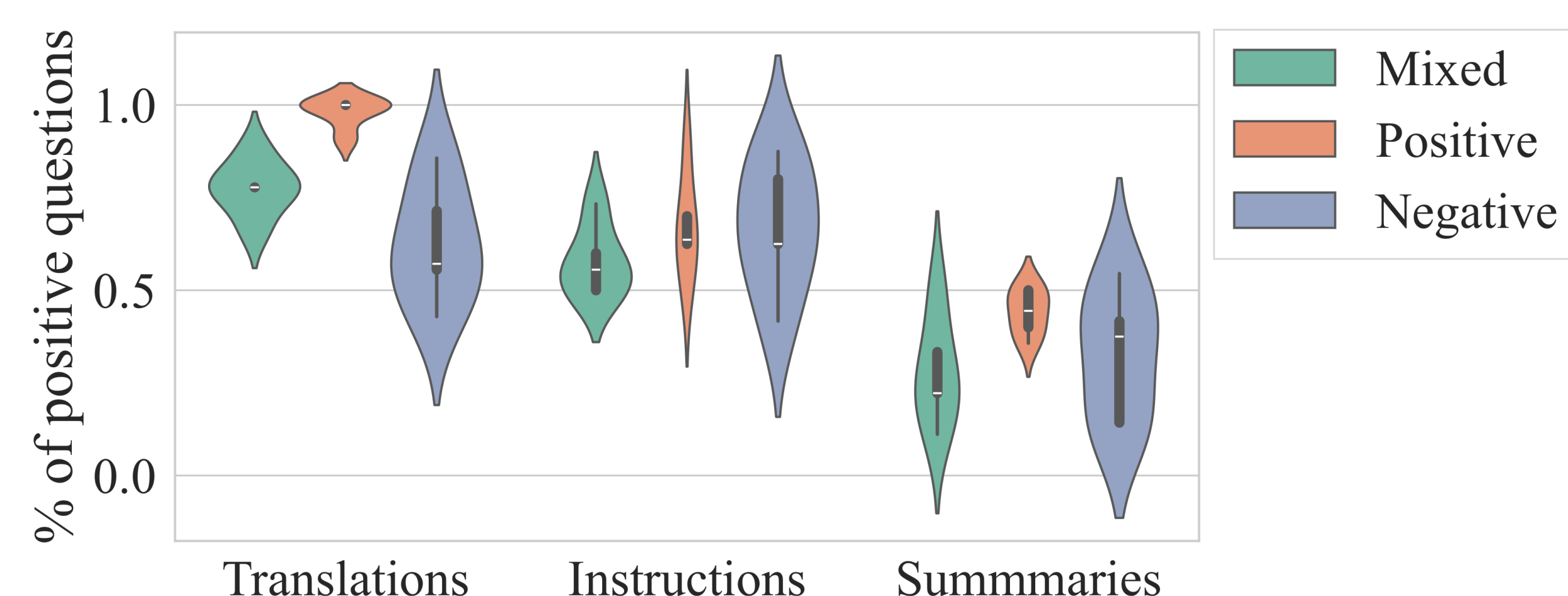
— **Random wins.**

A random sample converges the best and stays stable.

— **Skewness hurts.**

Best-only/worst-only data → narrows down to unstable questions.

RQ2 · ROBUST TO HOW YOU SEED IT



— **Convergence.**

A single-question seed already covers most properties.

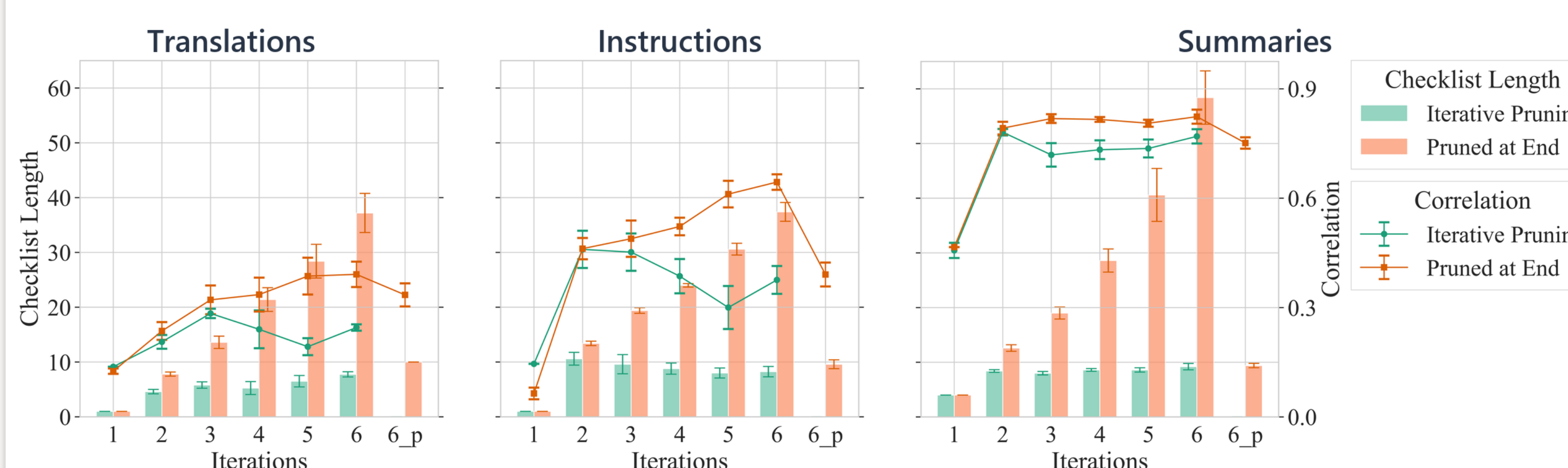
— **Tone.**

Positive / negative / mixed seeds → balanced, mixed tone.

— **Format.**

Yes/No & 1–5 scorecards reach near human correlation.

RQ3 · PRUNING TRADE-OFF



— **Stay lean.**

Iterative pruning keeps checklist tight (~10 Qs).

— **Coverage vs. quality.**

Prune-at-end broadens coverage; 7/8 properties shared.

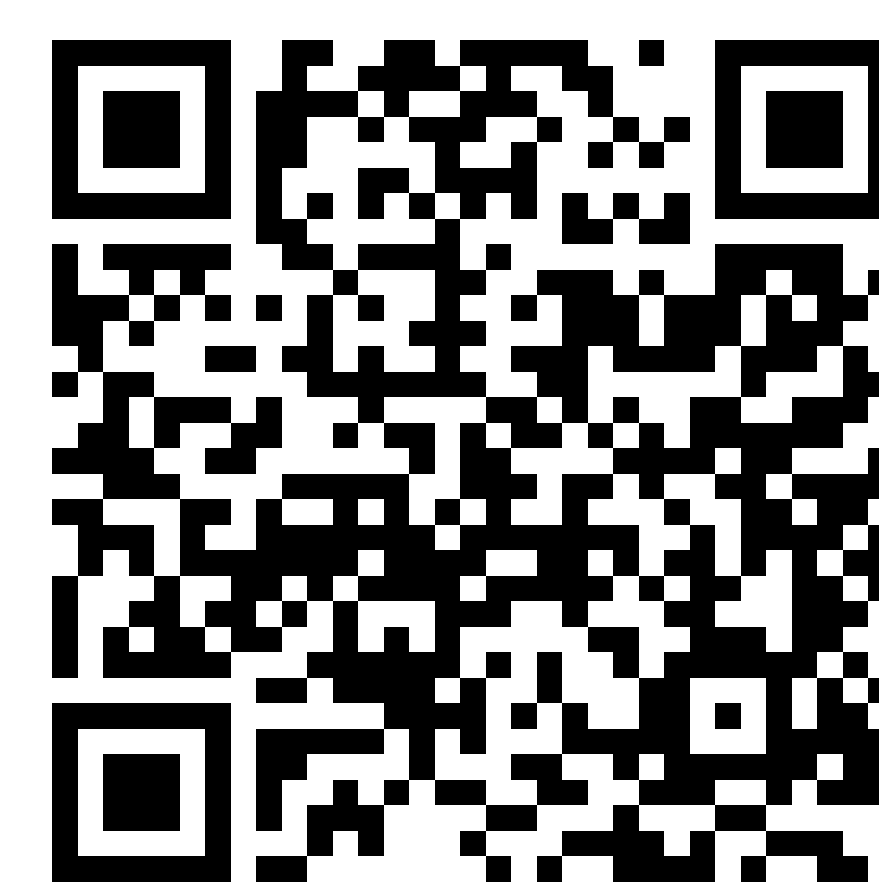
WHY IT MATTERS

- ✓ Easily deployable to build reliable evaluation pipelines.
- ✓ Subjective → verifiable concrete, checkable questions.
- ✓ Cuts manual effort of writing & maintaining checklists.
- ✓ Robust & reproducible across seeds, tones, formats, and datasets variations.

► Paper



► Code



AutoChecklist checklists that improve themselves!

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